Hieu Vo

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Education

Villanova University Expected May 2028

B.S. in Computer Engineering, Minor in Computer Science *GPA: 3.6/4.0*

- **Relevant Coursework:** Data Structures & Algorithms, Web & Mobile Dev, Database Systems, Computer Architecture, Operating Systems, Computer Networks, Network Security, Signal Processing
- **Honors/Awards:** Dean's List (All Terms), RevolutionUC Hackathon Winner – Best Use of MongoDB Atlas

Experience

Sharedi.io May 2026 – Present

Software Engineer Intern *Mountain View, CA*

- Architected a distributed notification engine using Java, Spring Boot and CockroachDB, supporting **100K concurrent users** while delivering real-time in-app updates via WebSocket with STOMP and email.
- Reduced RabbitMQ event payloads from 6 KB to 400 bytes by publishing only IDs and metadata, while batch-loading subscriber preferences to cut **1,001 database queries to 2** for 1,000 recipients.
- Engineered a distributed Redis caching layer for unread notification aggregates, cutting read latencies from **20–50 ms to under 5 ms** under high concurrency.
- Automated deployment of the **Spring Boot** service to **Google Compute Engine** and added **HMAC authentication**, idempotent event processing, and service health checks.

Media Landscape (MeLa) Lab Oct. 2025 – May 2026

Software Engineer Intern *Villanova, PA*

- Built a Django REST and PostgreSQL analytics platform over **40M** news articles, enabling researchers to compare media bias across 500+ outlets; optimized database indexes to reduce query latency from 2.5 to 1.5 seconds.
- Implemented a Kafka-based NLP pipeline with parallel Python workers, increasing ingestion from **15K to 60K articles/day** and reducing batch processing time from 5 hours to 1 hour.
- Developed 10 automated **Python** data pipelines to process 60K articles daily from 500 outlets, transforming raw article data into structured features exposed through 10 REST API endpoints.
- Built a **BERTopic**-based processing pipeline over 50K articles to identify major news themes and support automated topic assignment across a 40M-article corpus for downstream analytics.

L3Harris Technologies Jun. 2025 – Jul. 2025

Software Engineer Intern *Bryn Mawr, PA*

- Engineered a high-throughput image-processing workflow using Python and OpenCV to convert 4K wind tunnel videos into **12K paired training images** for an AI-based airfoil screening system.
- Optimized a GPU-based model training pipeline through adaptive batch sizing and memory management, reducing training time from 10 hours to 2 hours.
- Developed a containerized aerodynamic screening service using **FastAPI**, **MySQL**, and Docker, exposing model predictions through backend APIs for on-demand airfoil evaluation.

Projects

AI Coding Memory System | *Go, PostgreSQL, pgx, Docker Compose, golang-migrate*

- Implemented a persistent **Go** daemon and **PostgreSQL** backend for cross-session AI coding memory, using pgx connection pooling to serve multiple coding-tool clients without per-command database initialization.
- Built PostgreSQL full-text search with weighted indexing and relevance ranking, achieving **sub-second latency** while capping retrieved context at 2,000 tokens to reduce AI context overhead.
- Reduced local setup to **under 5 minutes** by automating PostgreSQL setup, Docker Compose startup, and database migrations, while keeping the system local with generated credentials and secure config permissions.

Clinical Trial Matching Platform | *Python, FastAPI, MongoDB, Gemini API, LangChain, LangGraph, ChromaDB*

- Architected a clinical trial recruitment platform using Python, FastAPI, MongoDB, and React.js, orchestrating multi-agent AI workflows that reduced patient–trial matching latency from **8 seconds to 4 seconds**.
- Moved OCR processing client-side with Tesseract.js, replacing **5 MB** patient-document uploads with **50 KB** extracted-text payloads while preventing raw documents from reaching the backend.
- Built a structured retrieval and reasoning pipeline using ChromaDB and LangChain, LangGraph with Gemini-powered agents, improving trial-matching accuracy from **65% to 78%** on patient-trial eligibility evaluations.

Technical Skills

Languages: Python, Java, Go, C, C++, JavaScript, TypeScript, SQL
Frameworks & Libraries: Spring Boot, React, Django REST, FastAPI, PyTorch, OpenCV, LangChain, LangGraph
Systems & Databases: PostgreSQL, CockroachDB, MongoDB, Redis, Kafka, RabbitMQ, Docker, GCE, Linux, Git